Experiment plan

RQ1: How does debiasing work? Or how does safety alignment work?

- Does the bias not emerge in the model or does the bias appear and is later repressed in later layers?

RQ2: How does jailbreaking work?

- Does the model reuse the old bias pathways?
- Does the model uncover new pathways for bias?
- Is this the same for different jailbreaking strategies?
- Is there any regression on performance if both the old and new pathways are edited?

$\mathcal{M}$ → A biased model

P_S → Safety prompt

P_J → Jailbreaking prompt

x_i→ safe input (say a pro-stereotypical example)

x_k → unsafe input (say an anti-stereotypical example)

We denote biased input with no prompt, safety prompt, and jailbreaking prompt as x_i^B, x_i^S, x_i^J respectively.

Experiment 1) Where does bias emerge?

$\mathcal{M}(x_k^B), \mathcal{M}_{h_{l,T} \leftarrow h_{l,T}^{iB}}(x_k^B)$

Experiment 2) Where does debiasing happen?

$\mathcal{M}(x_k^S), \mathcal{M}_{h_{l,T} \leftarrow h_{l,T}^{kB}}(x_k^S)$

Experiment 3) Where does jailbreaking happen?

$\mathcal{M}(x_k^J), \mathcal{M}_{h_{l,T} \leftarrow h_{l,T}^{kS}}(x_k^J)$

There is one concern though. Adding safety and jailbreaking prompts naturally change the size of the inputs, potentially causing undesired effects in the model. So, instead what we can do it determine a full safety and jailbreaking prompt and then add noise to the prompt and jailbreaking embeddings for getting runs without any additional prompt and runs without jailbreaking prompt.

Dataset: WinoBias data

https://uclanlp.github.io/corefBias/overview

- x_A: The mechanic told nurse that he has not repaired the car yet. In this sentence, he refers to the
- x_B: The mechanic told nurse that she has not repaired the car yet. In this sentence, she refers to the

sc(target|x_A)

sc(target|x_B)

$sc(target|x_B, h_{l,T}^{(A)})$